

ECE498A: Engineering Design Project

Welcome!

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May 26, 2026

Today's Agenda

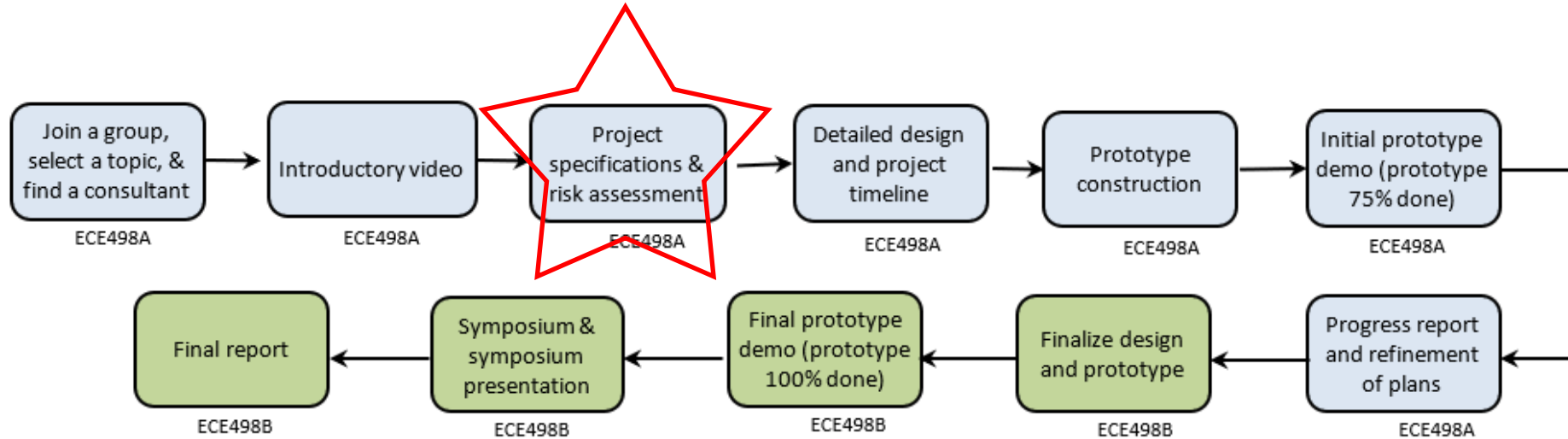
1. Course Deliverables and Grading
2. Expectations for the Project Specifications and Risk Assessment Deliverable
3. Safety Quiz and Safety

Course Deliverables and Grading

Deliverable	Dates and Locations	Grader	More information	Weight of Final Course Grade
Project Abstract & Project Agreement Form*	Uploaded to your LEARN Dropbox by 11:59pm on May 31	Course instructors	Page 10	10% (8+2)
Safety Quiz	Must be done by 11:59pm on May 31 (LEARN)	Graded automatically by LEARN	Page 9	5%
Introductory video	Must be done by 11:59pm on May 31 (LEARN)	Course instructors	Page 9	5%
Project Specs. & Risk Assessment Document*	Uploaded to your LEARN Dropbox by 11:59pm on June 7	Course instructors	Page 15	15%
Detailed Design and Project Timeline Document*	Uploaded to your LEARN Dropbox by 11:59pm on July 05	Course instructors	Page 18	35%
Initial Prototype Demonstration*	Suggested time is July 21--26. Must be done before 5:00pm on July 28	Project consultant (fills in Initial Prototype Demonstration Feedback Sheet)	Page 22	Part of Progress Report grade
Progress Report and Initial Prototype Demonstration Feedback Sheet*	Uploaded to your LEARN Dropbox by 11:59pm on Aug 02	Course instructors	Page 22 Page 24	30%
Revised Project Title and Abstract	Uploaded to your LEARN Dropbox by 11:59pm on Aug 02	Course instructors		-5% - 0%
Reimbursement paperwork	Uploaded to your LEARN Dropbox by 11:59pm Aug 02	N.A.	Page 34	0%

2. Expectations for the Project Specifications and Risk Assessment Deliverable

Project Specs and Risk Assessment Doc



Overview

The *Project Specifications and Risk Assessment* document is a stand-alone technical report that describes your project specifications and assesses the risks that may jeopardize the success of your project.

Project Specs and Risk Assessment Doc

Writing Style

- This is a stand-alone formal technical document.
- See {14} for requirements on units, figures, tables, captions.
- Minimize jargon and explain specialized concepts.

Page Limit: 10 pages, including everything

Font size, spacing, margins: See page {14}.

Project Specs and Risk Assessment Doc

Required Components

Title page (see {15} for details)

- Include your group number (e.g., 2026.75), student details (names, IDs, email addresses), name of consultant.

Table of Contents

Project Specs and Risk Assessment Doc

Required Components

Section 1: High-Level Description of Project

1.1 Motivation

- Expand on the motivation from your Project Abstract.
- Include at least one reference to justify motivating facts.
- Include figures if that will help to demonstrate your project motivation.

Project Specs and Risk Assessment Doc

Required Components

Section 1: High-Level Description of Project

1.2 Project Objective

- Must be clear and explicit:
 - “The objective of this project is to design ... to achieve ...”
- Take from your Project Abstract and revise as needed.
- Include figures if that helps to explain the objective more clearly.

Project Specs and Risk Assessment Doc

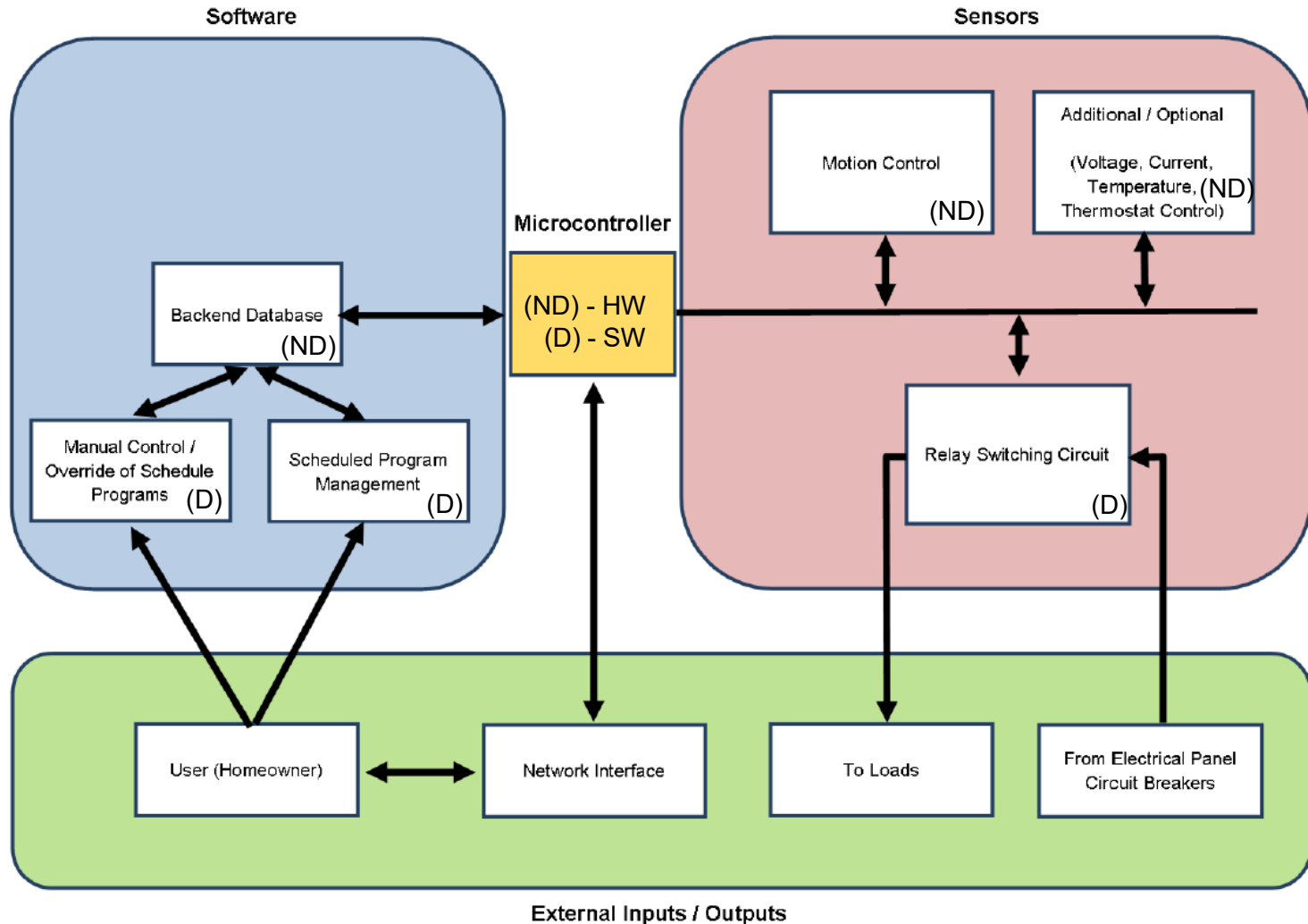
Required Components

Section 1: High-Level Description of Project

1.3 Block Diagram

- A block diagram provides a “big picture” description of your project.
- It should be well labelled, showing the project’s inputs, subsystems, and outputs. Aim for 4-6 subsystems.
- Each subsystem is associated with either:
 - Something you will be designing
 - Something that is already chosen (i.e., although part of your project, it is not something you’ll be designing)

Example (Power Management System)



Project Specs and Risk Assessment Doc

Required Components

Section 2: Project Specifications

- A project specification tells the designer what the requirements are for the final system.
- Project specifications fall into two categories:
 - **Functional** specifications
 - **Non-functional** specifications

Project Specs and Risk Assessment Doc

Required Components

Section 2: Project Specifications

2.1 Functional Specifications

- A functional spec deals directly with what the system is supposed to do (i.e., its operation or behavior).

Example: “The quadcopter must land autonomously within 5 seconds of a low-battery warning.”

Note: Assign each functional specification an ID (e.g., FS1, FS2, ...) and refer to it in the appropriate places: verification, risks, design, ...

Project Specs and Risk Assessment Doc

Required Components

Section 2: Project Specifications

2.2 Non-functional Specifications

- A non-functional design spec deals with everything else (typically reliability, usability, size, mass, appearance,, ...).

Example: “The designed control system for the quadcopter must fit within a 3 cm by 5 cm by 5 cm box.”

Note: Assign each non-functional specification an ID (e.g., NFS1, NFS2, ...) and refer to it in the appropriate places: verification, risks, design, ...

Project Specs and Risk Assessment Doc

Required Components

Section 2: Project Specifications

For your document:

- Write the specs in table form with additional text as needed.
- Label each specification as essential or non-essential.

Essential: those that your project must satisfy to be considered satisfactory.

Non-essential: less critical, but by satisfying them your project will be better than just satisfactory.

Project Specs and Risk Assessment Doc

Required Components

Section 2: Project Specifications

For your document:

- Write the specs in table form with additional text as needed.
- Label each specification as essential or non-essential.
- Each spec must be verifiable.

Project Specs and Risk Assessment Doc

Required Components

Section 2: Project Specifications

For your document:

- Write the specs in table form with additional text as needed.
- Label each specification as essential or non-essential.
- Each spec must be verifiable.

Verifiable: Someone assessing your project can objectively determine whether or not the spec is satisfied.

Project Specs and Risk Assessment Doc

Required Components

Section 2: Project Specifications

For your document:

- Write the specs in table form with additional text as needed
- Label each specification as essential or non-essential
- Each spec must be verifiable
- Each spec is appropriate

Project Specs and Risk Assessment Doc

Required Components

Section 2: Project Specifications

For your document:

- Write the specs in table form with additional text as needed.
- Label each specification as essential or non-essential.
- Each spec must be verifiable.
- Each spec is appropriate.

Appropriate: Do not include irrelevant specs.
Do not include design decisions.

Project Specs and Risk Assessment Doc

Required Components

Section 2: Project Specifications

For your document:

- Write the specs in table form with additional text as needed.
- Label each specification as essential or non-essential.
- Each spec must be verifiable.
- Each spec is appropriate.
- The set of specs is complete.

Project Specs and Risk Assessment Doc

Required Components

Section 2: Project Specifications

For your document:

- Write the specs in table form with additional text as needed
- Label each specification as essential or non-essential
- Each spec must be verifiable
- Each spec is appropriate
- The set of specs is complete

Complete: All the important aspects of the system behaviour are captured by the specifications.

Project Specs and Risk Assessment Doc

Required Components

Section 2: Project Specifications

For your document:

- Write the specs in table form with additional text as needed.
- Label each specification as essential or non-essential.
- Each spec must be verifiable.
- Each spec is appropriate.
- The set of specs is complete.

Complete: Also, each subsystem you will be designing has at least one associated specification.

Project Specs and Risk Assessment Doc

Required Components

Section 2: Project Specifications

For your document:

- Write the specs in table form with additional text as needed.
- Label each specification as essential or non-essential.
- Each spec must be verifiable.
- Each spec is appropriate.
- The set of specs is complete.
- Specs are open-ended.

Project Specs and Risk Assessment Doc

Required Components

Section 2: Project Specifications

For your document:

- Write the specs in table form with additional text as needed.
- Label each specification as essential or non-essential.
- Each spec must be verifiable.
- Each spec is appropriate.
- The set of specs is complete.
- Specs are open-ended.

Open-ended: The spec may be designed/implemented in several ways.

Project Specs and Risk Assessment Doc

Required Components

Section 2: Project Specifications

For your document:

- Write the specs in table form with additional text as needed.
- Label each specification as essential or non-essential.
- Each spec must be verifiable.
- Each spec is appropriate.
- The set of specs is complete.
- Specs are open-ended.
- At least some of the specs are **challenging** for a group of fourth-year engineering students.

Project Specs and Risk Assessment Doc

Required Components

Section 3: Risk Assessment

- Identify at least 3 situations that could prevent successful completion of your project
- For each situation, specify in table form:
 - The nature of the situation

Examples: insufficient skill or knowledge, technical risks such as insufficient battery power or processing power, ...

Project Specs and Risk Assessment Doc

Required Components

Section 3: Risk Assessment

- Identify at least 3 situations that could prevent successful completion of your project
- For each situation, specify in table form:
 - The nature of the situation
 - The potential impact of the situation

Specify as high, medium, or low impact

Project Specs and Risk Assessment Doc

Required Components

Section 3: Risk Assessment

- Identify at least 3 situations that could prevent successful completion of your project.
- For each situation, specify in table form:
 - The nature of the situation
 - The potential impact of the situation
 - The probability of the situation occurring

Specify as high, medium, or low probability

Project Specs and Risk Assessment Doc

Required Components

Section 3: Risk Assessment

- For situations with high impact and/or high probability, specify:
 - What can be done to reduce the likelihood that the situation occurs?
 - What can be done to reduce damage/harm if the situation does occur?

Project Specs and Risk Assessment Doc

Required Components

References

- Use IEEE formatting, and number references as [1], [2], etc.
- Be sure to properly reference any figures taken from other sources.

Example:

[1] T. Khalifa, A. Abdrabou, K. Naik et al., “Split- and Aggregated-Transmission Control Protocol (SA-TCP) for Smart Power Grids,” *IEEE Transactions on Smart Grids*, Vol. 5(1), 2014, pp. 381-391.

Project Specs and Risk Assessment Doc

Grading

- This deliverable is worth 20%, marked out of 50:

Item	Marks
Presentation and layout	06
Block diagram	12
Specifications	26
Risk assessment	06

3. Safety Quiz and Safety

Safety Quiz

Safety Quiz

- Each student must complete an on-line safety quiz through LEARN. Deadline: See page {8} of course outline.
- 15 questions are randomly chosen
- You get 3 attempts, and the highest score is taken
- Worth 5% of your final grade in ECE498A.
- Questions are selected from these topic areas:
 - E2-3353 safety rules (from *Safety Rules for E2-3353*)
 - Electric & mech. safety (from *Safety Rules for E2-3353*)
 - Basic electrical calculations (easily found on the web)
 - Safety in general (regulations, policies) {5}
 - University Safety Office {5}
 - Symposium safety {6}

Safety Quiz

- Aids allowed:
 - During the quiz, you **may not consult with another** person (directly or indirectly).
 - During the quiz, you **may consult any document**, including anything you find on the internet, to help you.
 - In fact, you are expected to **look up information on the internet** to correctly answer many of the questions.

Safety

Safety

- Safety is important in this course because you will be constructing prototypes. **Things** like motors, circuits, soldering, and tools **can be dangerous** if not handled correctly.
- Next March you will be demonstrating your prototypes in a **public symposium**, so safety is doubly important there.

Safety

University Safety Office

See page 5 for the URL.

They have on-line resources and short training sessions.

Online	Course Description	Course Number	Approximate Duration
Online	Biosafety	SO1069	1.5 hours
Online	Cryogenic and compressed gas safety	SO1030	45 minutes
Online	Laboratory safety	SO1010	45 minutes
Online	Laser safety training theory	SO1066	2 hours
Online	Radiation safety open sources	SO1013	3 hours
Online	Radiation safety devices	SO1017	45 minutes
Online	Radiation safety sealed sources	SO1015	2 hours
Online	Radiation safety transportation	SO1021	2 hours
Online	Supervisor's safety awareness	SO1100	1 hour
Online	WHMIS for employees	SO1002	1 hour
Online	Working in clean rooms	-----	2 hours
Online	X-ray safety	SO1011	2 hours

Safety in General

A. University employees and students are required to work in a safe manner and to encourage others to work in a safe manner. Why?

1. Those who show reckless disregard for safety of others are criminally negligent under the Criminal Code of Canada
2. Failure to correct dangerous situations is an example of professional misconduct for Ontario Engineers.
3. Canadian and Ontario occupational health and safety legislation applies to all of us.
4. Policy 34 of the University requires all employees and students to adhere to safety legislation/procedures.

Safety in General

B. It's impossible to make a workplace 100% free of hazards.

A key goal of engineering is to reduce the risk of injury or illness to an acceptable level.

Consequently, you can never be casual about safety. Think about safety as you work in the lab, as you construct your prototype, as you test it out, and as you demonstrate it to others.

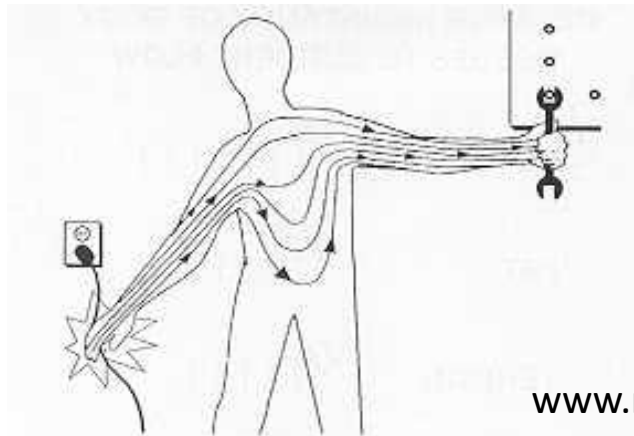
Electrical Safety

- C. Electrical hazards are especially relevant for most of your projects
1. **Do not overload** an electrical system—overloading produces heat or arcing. **Use** proper gauge wiring. **Install** a fuse to limit current. **Avoid** metal jewelry, pens, and rulers.



Electrical Safety

- C. Electrical hazards are especially relevant for most of your projects
- 2. **Don't underestimate** how dangerous a shock hazard can be. As little as **10 mA** flowing through a muscle will cause that muscle to contract—you literally cannot let go of the wire! **100 mA** can be fatal.



www.medbc.com

Electrical Safety

C. Electrical hazards are especially relevant for most of your projects

3. If your project includes

- a. Custom 120V AC circuitry, or
- b. Any other circuits running at voltages above 40V, or
- c. Any circuit running at currents greater than 3A, or
- d. Any size of lead-acid battery

then before starting to build your circuit you must talk to you consultant about incorporating safety into circuit design...

4. ... and you need to work in pairs whenever you are probing or modifying your circuit.

Electrical Safety

- C. Electrical hazards are especially relevant for most of your projects **(7 best practices)**
- a) Circuits should be **enclosed in a plastic or metal box** that is approved for electrical use.



forum.arduino.cc



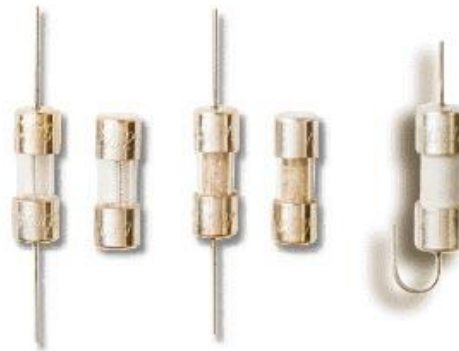
Photo by Douglas Harder

Electrical Safety

- C. Electrical hazards are especially relevant for most of your projects **(7 best practices)**
- a) Circuits should be **enclosed in a plastic or metal box** that is approved for electrical use.
 - b) Circuits should have appropriate **power switches and fuses**.



www.eloybecker.com



www.thefusecompany.net

Electrical Safety

C. Electrical hazards are especially relevant for most of your projects (7 best practices)

- a) Circuits should be enclosed in a plastic or metal box that is approved for electrical use.
- b) Circuits should have appropriate power switches and fuses.
- c) High-capacity batteries should have well-insulated connections, overload protection, and a means of disconnection.



www.megabatteries.com



www.diytrade.com



www.telegraph21.com

Electrical Safety

C. Electrical hazards are especially relevant for most of your projects **(7 best practices)**

d) **Proper wires** and connectors should be used.

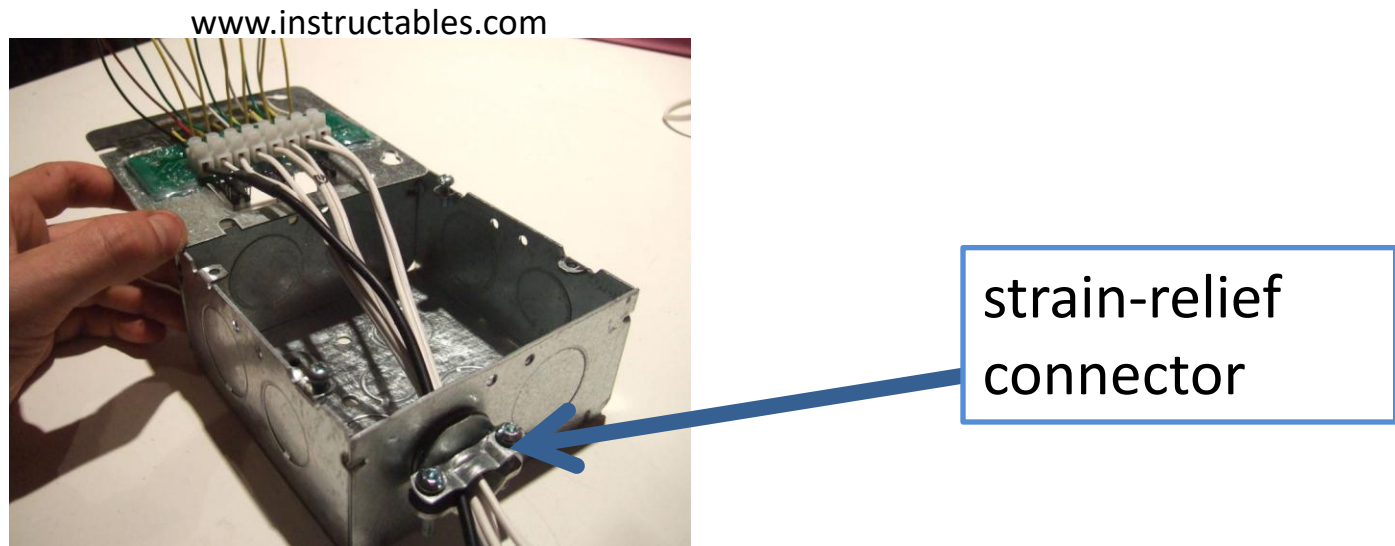


www.photo-dictionary.com

Electrical Safety

C. Electrical hazards are especially relevant for most of your projects **(7 best practices)**

- d) **Proper wires** and connectors should be used.
- e) Power cords entering electrical boxes should have an approved means of **strain relief**.



Electrical Safety

C. Electrical hazards are especially relevant for most of your projects (7 best practices)

- d) Proper wires and connectors should be used.
- e) Power cords entering electrical boxes should have an approved means of **strain relief**.
- f) All 120V AC devices, including power adapters, should be **approved by CSA, UL, ULC, or ESA**.*

**See warning on {6} about CE approval being insufficient*



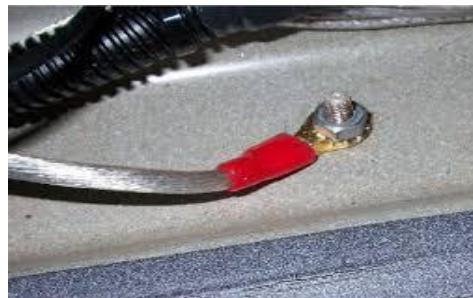
www.dlyaudio.com

CSA: Canadian Standards Association
ESA: Electrical Safety Authority
UL: Underwriters Labs (Safety Consulting + Cert Co.)
ULC: UL of Canada
CE: European Conformity

Electrical Safety

C. Electrical hazards are especially relevant for most of your projects **(7 best practices)**

- d) **Proper wires** and connectors should be used.
- e) Power cords entering electrical boxes should have an approved means of **strain relief**.
- f) All 120V AC devices, including power adapters, should be **approved by CSA**, UL, ULC, or ESA.
- g) All circuits should be appropriately **grounded**.



Electrical Safety

- C. **Electrical hazards are especially relevant for most of your projects**
- 6. Study the document *Safety Rules for E2-3353* on LEARN (and posted in E2-3353) for more electrical safety info.

For even more info on electrical safety, see the excellent slide deck on LEARN called “Electrical Safety”. The slides were produced by Minerva, a safety management education not-for-profit corporation.

Mechanical Safety

D. Mechanical hazards are relevant for some projects

1. You need to **build** appropriate **mechanical shielding** into your prototype if your project involves:
 - a. Moving parts that may pinch, hit, or crush a person, or
 - b. Sharp edges or points, or
 - c. High-pressure gases or liquids.



Mechanical Safety

D. Mechanical hazards are relevant for some projects

2. You need to incorporate bracing into any part or structure that risks falling over.



Safety Rules for E2-3353

E2-3353 is the ECE project lab room, open to all ECE students as a resource to help with open-ended projects. See {4}.

We'll discuss use of this room more at a future session, but now let's focus on safety in the room. Details are in the document *Safety Rules for E2-3353* on LEARN and posted in the room.

Safety Rules for E2-3353

General

1. Safety is an important consideration when working on projects. Safety is everyone's responsibility.
2. There is a first-aid station in the project room, right next to the soldering stations. Other nearby first-aid stations are in E2-3349 and E2-2336.
3. Emergency contacts are posted behind the door and on the safety notice board.

Safety Rules for E2-3353

Soldering

1. All soldering must be done at a **soldering station**.
2. **Safety goggles** must be worn to prevent possible eye injuries caused by splattering hot solder when soldering.
3. A **fume extractor** must be used to prevent the inhalation of hazardous fumes when soldering.
4. Clean up **debris** (wire ends, etc.) after you have finished soldering.
5. New soldering tips may be obtained from the Lab Instructor.



Safety Rules for E2-3353

Clothing, Tidiness

1. **Shoes** must be worn at all times to prevent injuries to feet caused by heavy objects, sharp objects, chemicals, and other lab hazards when working. No flip-flops or open-toed sandals are allowed in the design project laboratory.
2. Do not wear **loose clothing**.
3. If you are working with circuits, do not wear any **metal jewelry**.
4. If you are working with moving parts (motors, gears, pulleys, etc.) then **tie back long hair**.
5. Keep the work area and workbench as **tidy** as possible and clear of items not needed.

Safety Rules for E2-3353

Electrical Shock Prevention

6. When making electrical measurements, **use only one hand** at a time to avoid accidentally creating a circuit. No part of a live circuit should be touched by a bare hand.
7. **Remove** all loose conductive jewellery and fashion items, including rings and bracelets.
8. **No ungrounded** electrical or electronic apparatus is to be used in the project room unless the apparatus is double insulated or battery operated.



www.pinterest.com

Safety Rules for E2-3353

Electrical Shock Prevention

9. “Cheater cords” and 3-to-2 prong 120 V AC adapters are prohibited as is any other method used to disable the equipment ground (via the round pin on an AC power plug) on electrical instruments.
10. Make sure the **power switch is OFF** before plugging an electrical device into an outlet and do not stand directly over the Device Under Test (DUT) when the power is turned on. Also, turn instruments or equipment OFF before unplugging from the 120 V AC supply.
11. Consider all circuits to be "hot" (**live**) unless proven otherwise.

Safety Rules for E2-3353

Electrical Shock Prevention

- 12. When unplugging a power cord, **pull on the plug** not on the cable.
- 13. When disassembling a circuit, first remove the source of electrical power.
- 14. All live circuits (including battery charging) must be supervised.



Safety Rules for E2-3353

Electrical Shock Prevention

15. Always consult the **instruction manual** provided by the manufacturer of the equipment. Information regarding safe use and possible hazards should be studied carefully.
16. Note where the **room electricity shut-off switch** is located.

Safety Rules for E2-3353

Fire Prevention

1. Always turn off the soldering iron when not in use.
2. No open flames are allowed in the project room.
3. Keep liquids, chemicals, and heat away from instruments and circuits.
4. Keep flammable materials away from sources of heat (e.g., soldering irons).

www.cbc.ca



Safety Rules for E2-3353

Drilling, Sanding, Filing, Painting

E2-3353 is not outfitted with any special ventilation systems or vacuum systems, so the following activities are not allowed in the room:

- Drilling
- Sanding
- Filing
- Painting
- Use of solvents/aerosols



Safety Rules for E2-3353

Drilling, Sanding, Filing, Painting

Use the **student design shop in E5-1101D** for activities such as drilling, sanding, filing or painting materials.

Solvents or aerosols are best used outdoors.

You must follow the student shop safety rules and go through safety training (see <https://uwaterloo.ca/engineering-machine-shop/>).

Questions?